control systems training kits available in the laboratory, providing them with opportunities to experiment with different hardware and system dynamics. This lab-intensive course bridges the gap between theoretical concepts and practical implementation, equipping students with essential skills to design and deploy control solutions across a wide array of engineering applications. **Prerequisite**: MECA 440. Annually.

MECA 441	**Sensors and Actuators**	**3(3,0)**
Sensor and actuator technologies; classification and terminology of sensors and actuators; sensors characteristics; physical principles of sensing; measurement applications (position and speed, stress and strain, temperature, vibration and acceleration, pressure and flow…); Electromagnetic principles; solenoids and relays; electric motors (DC motors, Stepper motors…); Hydraulics actuators; Pneumatic actuators; interface methodologies and circuits; integration aspects; manufacturing techniques and material properties; team projects. **Prerequisite**: MECA 341. Annually.		

MECA 442	**Microcontrollers for Mechatronics**	**3(2,2)**
This course aims to provide a deep understanding of the integration of microcontrollers in mechatronic systems, starting from the technologies used to develop embedded systems to the internal features and peripherals of microcontrollers/microprocessors (timers, interrupts, analog-to-digital converters…). In this course, students develop various microcontroller programs for mechatronic applications, including the usage of I/O and communication peripherals. **Prerequisite**: CCEE 221 and MECH 200.		

MECA 443	**Mechatronic System Design**	**3(3,0)**
This course integrates all modules in mechatronics systems design. It consists of an overview of mechatronics; mechatronics systems design strategies; advanced modeling; use of control strategies; design of electric actuators; design of sensors; power electronics; hardware and software integration; and team projects on the design and implementation of integration between mechanical, electrical, and control systems. **Prerequisite**: MECA 440, 441, and 442.		

MECA 443L	**Mechatronic System Design Lab**	**1(0,2)**
Overview of mechatronic systems; integration of mechanical components, sensors, and actuators into mechatronics applications; use of switchgear components; integration of microcontrollers and FPGAs; LabVIEW programming; Design of power supplies. **Co-requisite**: MECA 443. Annually.		

MECA 444L	**Mechatronic Systems Programming**	**1(0,2)**
This introductory 1-credit course provides mechatronics students with a foundational understanding of Python. The curriculum is specifically designed to equip students with the essential Python skills necessary for subsequent coursework and applications in robotics, automation, and embedded systems. Emphasis will be placed on developing a strong grasp of core Python concepts, data structures, and control flow, laying the groundwork for using Python in the Robot Operating System (ROS), for artificial intelligence (AI) applications on platforms like Raspberry Pi and NVIDIA Jetson Nano,		

and for general scripting in mechatronic projects. Through hands-on exercises and practical examples relevant to mechatronics, students will gain the confidence to apply Python for tasks such as data acquisition, sensor interfacing, basic control algorithms, and interaction with robotic software frameworks. **Prerequisite**: CCEE 214 and MECH 200. Annually.

MECA 445	**Advanced Programming for AI & Robotics**	**3(3,0)**
This advanced course, built upon the prerequisite "Mechatronics Systems Programming," delves into the practical application of Python for developing intelligent robotic systems. A significant portion of the course will be dedicated to the Robot Operating System (ROS), providing students with the skills to design, implement, and test complex robotic software. Students will learn core ROS concepts, including nodes, topics, services, and parameters, as well as essential ROS tools for simulation and real-world robot interaction. Furthermore, the course will explore techniques for integrating Artificial Intelligence (AI) into mechatronics applications using Python. This will include an introduction to relevant Python libraries for machine learning, computer vision, and data processing, along with practical examples of how AI can be applied to tasks such as perception, decision-making, and control in robotic systems. The course will emphasize hands-on projects and assignments using platforms such as Raspberry Pi and NVIDIA Jetson Nano to provide students with practical experience in developing and deploying advanced robotic and AI-driven mechatronic solutions. **Prerequisite**: MECA 444L. Annually.		

MECA 499	**Co-op Work Experience**	**3(3,0)**
This Co-op work experience is designed to provide students with a five-week short-term opportunity in mechanical engineering. Students are encouraged to network in the discipline-related industries. A report, a poster, and a PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

MECA 541	**CAD/CAM/CAE**	**3(3,0)**
Fundamental principles of computer-aided design (CAD), computer-aided manufacturing (CAM), and computer-aided engineering (CAE); prototyping and distributed systems; specific rapid prototyping technologies; design for assembly; small batch manufacturing; continuous manufacturing; biotechnology and manufacturing. **Prerequisite**: MECH 224 and MECH 421. Annually.		

MECA 542	**Industrial and Manufacturing Control**	**3(2,2)**
This course will give an overview of programmable logic controllers (PLCs) and explores different PLC hardware components; number systems and codes; fundamentals of Logic; generation, loading and testing of PLC programs required to operate electrical motors and mechanical actuators used in industrial processes; developing fundamental PLC Wiring Diagrams and Ladder Logic Programs with an emphasis on programming timers and counters; integrated automation solutions through various programming architecture such as function-block and ladder diagrams; configuration of process visualization of a panel. Prerequisite: ELEC 320. On demand.		

MECA 544	Robotics	3(3,0)
Introductory historical development of robotics; static grasp analysis; rigid-object kinematics; degrees of freedom; robot arm kinematics; inverse kinematics; dynamics; kinematic singularities; introduction to trajectory planning and control of manipulators; screw motions. **Prerequisite**: MECH 321. Annually.		

MECA 595A	BE Summative Learning Project I	1(1,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes a literature review, a scope of work, and a project proposal. **Prerequisites**: Senior Standing, ENGL 217. Annually.		

MECA 595B	BE Summative Learning Project II	3(3,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes literature review, design work, data collection, experimentation, data analysis, and technical writing. **Prerequisite**: MECA 595A. Annually.		

II. Elective Courses

Students may meet their technical electives requirements from the following selection of elective courses.

MECA 484	Civionics	3(3,0)
Applications of mechatronic techniques to civil construction. This course will provide engineers with the background necessary to optimize design techniques and understand infrastructure performance, behavior, and condition. The successful integration of intelligent sensing of innovative structures will allow civil structural engineers to expand the design envelope and introduce new design concepts, materials, and innovation in civil engineering. **Prerequisite**: Instructor's consent. On demand.		

MECA 485	Autotronics	3(3,0)
This course explores topics such as sensors and actuators in automotive systems, communication systems, the CAN protocol, electronic circuits, automotive standards, electronic suspension and steering, engine management, and on-board diagnostics. **Prerequisite**: MECA 440.		

MECA 498	Special Topics in Mechatronics Engineering	3(3,0)
Guided study and research on major-related topics. **Prerequisite**: Instructor's consent. On demand.		

MECA 515	Modern Control Systems	3(3,0)
In this course, the student learns how to use data and correlate it with physical theory to build state-space and control models; design, build, and test controllability,		

feedback, observability, and stability; and perform black-box modeling in both the time and frequency domains. **Prerequisite**: MECA 440. On demand.

MECA 543	**Fluid Power Systems**	**3(3,0)**
Introduction to hydraulic and pneumatic systems: study of basic hydraulic/pneumatic circuit components: pipes, valves, pumps, motors, filters, accumulators, sensors, and silencers; analysis/design of various hydraulic/pneumatic circuits such as high-low, regenerative, and synchronization circuits; use of Matlab/Simulink for simulation of fluid power system circuits. **Prerequisites**: MECH 333 or MECH 335. On demand.		

MECA 545	**Integrated Automation Lab**	**1(0,2)**
This lab explores PID control algorithms; digital controllers and their components; tuning methods and process control; diagnosis and error search; analog value processing; communication between two or more PLCs. Co-requisite: MECA 542. On demand.		

MECA 581	**Systems Engineering**	**3(3,0)**
A course with an interdisciplinary approach intended to enable the realization of successful systems. It focuses on defining customer needs and required functionality early in the development cycle, documenting requirements, then proceeding with design synthesis and system validation while considering the complete problem, including operations, performance, test, manufacturing, cost, and schedule. This course emphasizes the links between systems engineering and the fundamentals of decision theory, statistics, and optimization. Introduction to the latest commercially successful techniques in systems engineering. **Prerequisite**: Instructor's consent. On demand.		

MECA 586	**Mechatronics Drives**	**3(3,0)**
Switched-mode converter; steady-state voltages and currents of step-down, step-up, inverting, and other power converters; power semiconductor devices in a switching converter; discontinuous conduction mode; solve converters operating in DCM; implement transformer isolation in a dc-dc converter; design and analyze the feedback systems of switching regulators; Design Project. **Prerequisite**: ELEC 320. On demand. **Prerequisite**: ELEC 320		

MECA 597	**Advanced Topics in Mechatronics Engineering**	**3(3,0)**
This course is designed to enable students to study a specific advanced topic of interest, carefully selected from the field of mechatronics engineering. The content outline for such a topic is to be determined by the instructor and approved by the department Chair. **Prerequisite**: Instructor's consent. On demand.		

III. AI-Restricted Elective Courses

Students must choose at least one elective course (3 credits) from the following selection of AI-Restricted elective courses.

MECA 535	Expert Systems and Applied Intelligence	3(3,0)
This course explores the core principles of intelligent systems and their application in solving complex engineering and decision-making problems. Students will learn perception methods, reasoning techniques, and search strategies, as well as key AI tools such as neural networks, evolutionary computing, fuzzy logic, and intuitionistic fuzzy logic. The course also introduces intelligent programming using Prolog/python to model knowledge and build rule-based expert systems. Through practical examples, students will gain hands-on experience in developing AI-driven solutions for real-world applications. **Prerequisite**: MECA 444L.		

MECA 536	Applied Deep Learning for Intelligent Systems	3(3,0)
This course introduces the basics of computer vision and machine learning, focusing on their use in intelligent systems. Students will learn techniques for object detection, image understanding, and sensor data interpretation, using deep learning tools to build decision-making models. Hands-on practice with simulation platforms will allow students to design and test vision-based solutions in realistic scenarios. At the end of the course, participants will gain practical skills in combining perception, planning, and control, preparing them for applications in autonomous systems and other technologies. **Prerequisite**: MECA 444L.		

MECA 546	Smart Energy Management Systems	3(3,0)
This course delves into smart energy management systems (SEMS), focusing on versatile operating modes in solar photovoltaic (PV) and battery energy storage (BESS) systems. It equips mechatronics engineering students with the fundamental knowledge and practical skills required to design, implement, and optimize these crucial technologies for a sustainable future. Furthermore, the course discusses the potential of using generative AI for the design and development of smart energy management systems. A capstone case study is presented and analyzed. **Prerequisite**: MECA 442.		

MECH 522	Mechanism Synthesis: Classical Kinematics to AI-Driven Optimization	3(3,0)
This course introduces the theory and computational practice of mechanism synthesis, with emphasis on the design of four-bar linkages for path, motion, and function generation. Classical kinematic synthesis methods are integrated with numerical optimization techniques, including Newton–Raphson, least-squares, and gradient-based methods, as well as intelligent metaheuristic algorithms such as genetic algorithms, tabu search, and ant colony optimization. The course also exposes students to data-driven and AI-assisted approaches, including deep neural networks for inverse mechanism design, as advanced computational tools for automation, design-space exploration, and decision support in mechanism synthesis. Computational projects provide students with hands-on experience in mechanism modeling, simulation, error analysis, optimization, and the synthesis of mechanical systems under practical design constraints. **Prerequisite:** MECH 321.		